Code No. 3309

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) III-Semester (MAIN) Examinations DEC-2025/ JAN-2026

PRINCIPLES OF PROGRAMMING LANGUAGES

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	CO
1.	a)	Differentiate general purpose languages and special purpose languages.	2	1	1

	b)	Consider the following C program:
		
		int fun(int _ i)
		{
		*i+=5;
		return 4;
		}
		void main
		{
		int x=3;
		x=x+fun (&x)
		}
		
		What is the value of x after assignment statement in main method assuming i. operands are evaluated left to right?
			2	2	2

	c)	Write a program to represent actual parameters and formal parameters.	2	2	3
	d)	What is the primary problem with semaphores to provide synchronization?	2	2	4
	e)	What are antecedents and consequents? Give an example.	2	2	5
	f)	Define grammar, derivation and a parse tree.	2	1	1
2.	a)	Explain the process of compilation in each phase of a compiler.	8	2	1
	b)	Construct an ambiguous grammar and derive the parse tree for A = B+C*A.	4	3	1

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3. a)	Write a short note on Data types with an appropriate example for each.	8	2	2
b)	Explain the structure of an associative array.	4	5	2
4. a)	What are the guarded commands? Draw the flow graph of dijkstra's selector and loop statement.	10	2	
b)	List out the design issues of multiple selection statements.	2	2	
5. a)	Define Exception. How do you implement exception handling in java? Explain with suitable example.	6	2	
b)	Demonstrate event handling in C++ and Java.	6	3	
6. a)	Compare and contrast functional programming languages and imperative programming languages.	6	3	
b)	Define Cambridge polish notation and write a LISP program function that calculates sum of numbers using a vector.	6	4	
7. a)	Explain about parameter passing methods with an example.	8	2	
b)	Give some reasons why computer scientists and professional software developers should study general concepts of language design and evaluation	4	3	

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